

Chapter 20

The Continual Simulation Report

Identity, Scale, and the Kernel Event Spine

IKK End-Tag Enrichment (WO-75A) · IKK Identity Vector (WO-75B) · v5.13
Freeze Summary

2026-06-28 | Commit: 489d11cd | 167/167 tests passing

Branch freeze notice. This document is the formal closure artefact for the v5.13 arc of VSEPR-SIM. The branch `feature/wizard-full-module-expansion` is frozen at this state. All deferred items are tracked in `STAGE.md` and `CLOSEOUT.md`. Work resumes on a new branch.

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20.0. Preamble

This chapter documents a single organising idea that runs through the entire v5.13 arc and gives every module within it its architectural justification: *the simulation must continuously print what it knows*.

The idea is not about logging. It is not a debugging aid. It is the foundational output doctrine of the platform: every event the kernel observes, every projection loss it incurs, every identity fraction it retains or discards—all of it must be expressed continuously, per step, in structured, human-readable, machine-parseable form.

This principle existed implicitly from Day 1 in the design of the `KernelEventLog`. The v5.13 arc formalised it, extended it upward into the IKK theoretical layer, and delivered two concrete implementations: the IKK End-Tag system (WO-75A) and the IKK Identity Vector (WO-75B).

20.1. The Continual Printing System

A. Doctrine

Core doctrine. A simulation that only produces a final result is not a simulation. It is a calculator. A simulation must continuously narrate its own state, uncertainties, and information losses across every step, every scale, and every event channel—so that the researcher can reconstruct not only *what happened* but *what the kernel knew*, and *what it could not know*, at every moment of the run.

This doctrine has three consequences for the architecture:

1. **No silent discards.** Every event the kernel processes is emitted—to the console, to the JSONL archive, or to both. The event limiter may sample the console under load; the JSONL archive is never sampled.
2. **Structured streams.** Each physical channel emits a distinguishable, grep-able prefix. A researcher running a live session can follow a specific channel without instrumenting new code.
3. **Identity accounting.** The simulation does not only report *what happened*; it reports *how much of what happened is still recoverable* in the language of IKK distinguishability ($\mathfrak{D}$) and cross-scale efficiency (η_{ab}).

B. The Four Live Print Streams

The runtime emits four structured event channels to the terminal and the JSONL archive simultaneously. Each is individually grep-able and forms a separate column in the JSONL event archive:

Listing 1: Representative output from all four live print channels.

```
[PP]    step=184 t=0.184 idA=e_001 idB=pos_001 r=0.018 Ere1=0.382
        action=ANNIHILATE
[CHEM]  step=184 t=0.184 idA=C_004 idB=O_002 r=1.24 type=nonbond
        action=NONE
```

```
[DECAY] step=522 t=0.522 id=Pu_239_001 mode=alpha daughter=U235_001 dE
=0.052
[ANN] step=184 t=0.184 pair=e_001+pos_001 Ein=1.022 Eout=1.022
products=gamma_001,gamma_002
```

Prefix	Channel	Trigger
[PP]	Particle-particle interaction	Any pair interaction; <code>ANNIHILATE</code> on gate pass
[CHEM]	Chemistry bond / nonbond	Bond formation, breaking, or nonbond pair event
[DECAY]	Single-particle decay	Decay event (alpha, beta, fission)
[ANN]	Anti-pair annihilation	Triggered <code>AnnihilationEvent</code>

Table 1: Live print stream channels. All four are console-visible and archived in JSONL.

C. The `KernelEventLog`

The `KernelEventLog` is the runtime substrate of the continual printing system. It is the single source of truth for all downstream analysis, reporting, and dashboard records.

Principle 20.1.1. The `KernelEventLog` owns the simulation history. The `.xyz/.xyzFull` files own the atomistic ground truth. No other data structure may claim to represent either. These two concerns do not overlap.

The event log feeds the downstream pipeline in a single directed chain:

`KernelEventLog` → `FormationRecord` → `run_pipeline()` → `AnalysisRecord` → `ReportRecord` → D

The IKK analysis layer taps into this chain at the `AnalysisRecord` boundary:

`AnalysisRecord` $\xrightarrow{\text{WO-75A}}$ `IKKEndTag` $\xrightarrow{\text{render}}$ `ikk_end_tag_md`, `ikk_end_tag_tex`

20.2. IKK End-Tag Enrichment (WO-75A, Part A)

A. Motivation

Before WO-75A, the simulation’s identity metrics—projection loss, hidden residual, distinguishability fraction—were computed and stored in `IdentitySidecarRecord` but never *explained*. Reports read as data dumps. The IKK End-Tag closes every report section with a structured block that interprets these numbers in the language of IKK theory (IKK I–IV, NOTATION REGISTRY v1.0).

B. The End-Tag Block Structure

Each end-tag block emits the following fields (Markdown and $\text{\LaTeX}$ variants):

Field	Symbol	Meaning
Scale regime	S_n	Atomistic / molecular / material regime
D mean	D_{rec}	Recoverable distinguishability fraction $\in [0, 1]$
D loss	$\Delta\mathfrak{D}$	Identity content lost across this run
Projection loss		Loss from atomistic $\rightarrow$ observable projection
Hidden residual	$ \Psi^{\text{hid}}\rangle$	Non-recoverable identity content estimate
Efficiency	η_{ab}	Cross-scale recovery fraction $\in [0, 1]$
Entropy delta	ΔS	Proportional to $-\Delta\mathfrak{D}$ (IKK 2nd law)
Formation memory		Fraction of formation history still recoverable
Second-law check		PASS / WARN / FAIL

Table 2: IKK End-Tag fields. All values are derived from `IdentitySidecarRecord`—never written back into particle state or `.xyz` truth files.

C. Second-Law Check

The second-law badge is a mandatory assertion appended to every section:

Definition 20.2.1 (IKK Second Law). $\Delta S > 0$ if and only if $\Delta\mathfrak{D} < 0$. Identity loss is monotonically paired with entropy increase in a non-reversible projection across scales.

Badge thresholds (configurable via `[analysis.ikk_end_tag]`):

- **PASS**: $D_{\text{rec}} \geq 0.60$ and second-law consistent.
- **WARN**: $0.35 \leq D_{\text{rec}} < 0.60$ or mild second-law tension.
- **FAIL**: $D_{\text{rec}} < 0.35$ or second-law violated.

D. Implementation Summary (Day 75)

Deliverable	File	Status
VsimIkkEndTagSection struct	include/vsim/vsim_document.hpp	DONE
apply_ikk_end_tag_key() parser wiring	src/vsim/vsim_parser.cpp	DONE
IKKEndTag struct + render functions	include/vsim/analysis/ikk_end_tag.hpp	DONE
IkkEndTagModule self-registering class	src/vsim/analysis/ikk_end_tag.cpp	DONE
AnalysisRecord output fields	include/vsim/analysis/i_analysis_module.hpp	DONE
Group 87 tests (20/20 PASS)	tests/test_ikk_end_tag.cpp	DONE
GL D-colour overlay (Part B, §20.3)	src/vis/renderer.cpp	DEFERRED

Table 3: WO-75A Part A deliverables. Part B is fully specified but deferred to the next arc.

E. VSIM Script Surface

Listing 2: `[analysis.ikk_end_tag]` block in a `.vsim` script.

```
[analysis.ikk_end_tag]
emit_markdown      = true
emit_latex         = false
d_pass_threshold   = 0.60
d_warn_threshold   = 0.35
scale_regime       = "S3"
section_reference  = "IKK IV sec 3.2"
```

20.3. WO-75A Part B — GL D-Colour Overlay (Deferred)

Deferred to next arc. The specification below is complete and frozen. Nothing has been removed from the design; only execution is deferred. The next arc begins here.

A. Colour Mapping Convention

The overlay maps $D_{\text{rec}} \in [0, 1]$ to a cold-to-hot palette following IKK convention (cold = identity preserved; hot = identity dissolved):

$$\begin{aligned} D_{\text{rec}} = 1.0 &\longrightarrow \text{deep blue} & D_{\text{rec}} = 0.75 &\longrightarrow \text{cyan} & D_{\text{rec}} = 0.5 &\longrightarrow \text{green} \\ D_{\text{rec}} = 0.25 &\longrightarrow \text{amber} & D_{\text{rec}} = 0.0 &\longrightarrow \text{red} \end{aligned}$$

B. Renderer Surface (Specified)

Listing 3: New renderer API surface – WO-75A Part B.

```
enum class RenderPass {
    STANDARD, DIST_OVERLAY, HIDDEN_RESIDUAL, SCALE_LADDER
};

void set_render_pass(RenderPass pass);
void set_dist_data(const std::vector<float>& per_atom_dist);
void set_hidden_residual_data(const std::vector<float>& per_atom_hid)
    ;
void render_dist_legend(int width, int height);
void render_scale_ladder_bar(int width, int height,
                             const std::string& active_regime
                             );
```

Toggle: D key cycles STANDARD → DIST_OVERLAY → HIDDEN_RESIDUAL → SCALE_LADDER → STANDARD.

C. Scale Ladder Panel

Listing 4: Scale ladder bottom bar concept.

```
S0(existence) --- S2(colour/EM) --- S3(spatial/EM) --- S4(weak) ---
S_mat
```

$$\eta_{\text{ab}} = 0.91 \quad |\Psi^{\text{hid}}| = 0.13 \quad \Delta\text{-D} = -0.11$$

20.4. IKK Identity Vector (WO-75B)

A. Overview

The IKK Identity Vector $\bar{i}_f$ is a five-component frame record that maps the simulation state at frame f into the IKK identity-information space:

$$\bar{i}_f = \begin{pmatrix} x \\ y \\ z \\ t \\ w \end{pmatrix} = \begin{pmatrix} \text{existence (presence)} \\ \text{EM / structural signature} \\ \text{spatial distinguishability} \\ \text{temporal coherence} \\ \text{internal / hidden channel} \end{pmatrix}$$

The vector is a *proxy* mapping in Phase 1 (v5.13.5); the full field-theory mapping is targeted for a future arc.

B. IKKIdentitySeries

Each run produces an `IKKIdentitySeries`: an ordered list of `IKKIdentityFrameRecord` entries, one per simulation frame. Each record carries $\bar{i}_f$, the frame-level $\Delta\mathfrak{D}$, and an entropy proxy.

Deliverable	File	Status
<code>IKKIdentityVector</code> + <code>IKKIdentitySeries</code>	<code>include/vsim/analysis/ikk_identity_vector.hpp</code>	DONE
<code>from_sidecar_record()</code> Phase 1 proxy mapping	same	DONE
<code>build_ivec_series()</code> + <code>write_identity_json()</code>	<code>src/vsim/analysis/ikk_identity_vector.cpp</code>	DONE
Group 88 tests (28/28 PASS)	<code>tests/test_ikk_identity_vector.cpp</code>	DONE
Phase 2 field-theory mapping	TBD	NEXT

Table 4: WO-75B deliverables at the v5.13 freeze.

C. VSIM Script Surface

Listing 5: `[analysis.ivec]` block in a `.vsim` script.

```
[analysis.ivec]
enabled          = true
write_identity_json = true
compute_ivec_flux = false    # J_iv -- identity-vector crossing
signature
```

20.5. D Time-Series Plot (WO-75A-C) — Delivered

`plot_dist_timeseries()` was added to `reporting/generate_report.py` and produces a dual-axis matplotlib figure embedded in every consolidated report:

- **Left axis:** D_{rec} over simulation steps (blue line).
- **Right axis:** entropy proxy ΔS (red dashed line).

- **Amber band:** region where $\Delta\mathfrak{D} < 0$ (identity loss zone).
- **Reference line:** $D = 0.5$ (IKK IV property activation threshold).

Output: `out/<run_id>/dist_timeseries.png`, embedded in `report.md` with a caption auto-generated from the IKK end-tag fields.

20.6. v5.13 Branch Freeze Summary

A. Test Registry at Freeze

Scope	Groups	Tests
Core / Geometry / VSEPR (Groups 1–22)	22	—
Pipeline / Dashboard / Parser (Groups 23–43)	21	—
EigenMine / CurveFit / Release (44–46)	3	10
CTL pipeline (Groups 58–69)	12	106
Dynx / DUB / Demo pipeline (71–77)	7	113
LengthScale / OutputFilter / GapClass (78–80)	3	—
ObserveMetrics / FormationHelper / Bio (81–83)	3	—
Replay / Discovery / CrystalMD (84–86)	3	—
IKK End-Tag (Group 87)	1	20
IKK Identity Vector (Group 88)	1	28
MCF-CAI (Group 89)	1	—
Total at freeze	89	167 / 167 PASS

Table 5: Test registry at the v5.13 freeze (Day 82, commit 489d11cd).

B. Arc Delivery Table

Version	Day	Deliverable
v5.0.0-beta.7	57	render_interval / step emission cadence
v5.0.0-beta.8	58	PBC / cell / boundary / Ewald
v5.0.0-beta.9	59	Registry resolution engine / CLI layer
v5.0.0-beta.10	60	Variance / N_evolution / while / batch sweep
v5.0.0-beta.11	61	Macro sampling / empirical verification
v5.0.0-beta.12	62	Batching upgrade / study orchestration
v5.1.13	72	Pillar F: empirical chemistry, Dynx, CTL pipeline
v5.13.3	73	Wizard modules 6–8 full field coverage
v5.13.4	74	Gap classifier + output filter + chemistry audit
v5.13.5	75	IKK End-Tag (WO-75A-A), Identity Vector (WO-75B), MCF-CAI (WO-76)
v5.13 FROZEN	82	Branch frozen. Chapter 20 authored as freeze artefact.

Table 6: v5.13 arc delivery table. Work commenced at Day 57 (v5.0.0-beta.7).

C. Deprecated at Freeze

ID	Item	Rationale
MF-F02	GitHub Actions CI workflow	Out of scope for v5.13. Not blocking any UC. Deferred indefinitely.

Table 7: Items deprecated at the v5.13 freeze.

D. Deferred to Next Arc

WO	Item	Next-arc entry point
WO-75A-B	IKK GL D-colour overlay	<code>src/vis/rendererer.cpp</code> : add <code>RENDER_PASS_DIST</code>
WO-75B gate	<code>vsepr doctor</code> full pass + tag push	Runtime data path audit
MF-B01	<code>[sweep]</code> runtime dispatch	<code>src/vsim/vsim_runtime.hpp</code> : sweep executor

Table 8: Items deferred to the next arc. All specifications are complete and frozen here.

Closing Note

The v5.13 arc set out to formalise a single idea that the project had been building toward since Day 1: that a simulation should not merely *compute* but *continuously account for what it knows and what it has lost*.

This chapter documents that arc’s architectural outcome. The Continual Printing System is now a named, documented doctrine. The IKK layer has been wired from theory (IKK I–IV) through schema (`VsimIkkEndTagSection`, `[analysis.ivec]`) into the runtime (`IkkEndTagModule`, `build_ivec_series()`), reporting (`render_ikk_end_tag_md()`, `plot_dist_timeseries()`), and test coverage (Groups 87–89, 48+ cases).

The GL overlay (Part B) is the one remaining visual expression of this system. Its specification is complete. The next arc begins there.

v5.13 arc — FROZEN | Day 82 | 167/167 tests | Commit 489d11cd
 Next arc opens from this state. First task: **WO-75A-B** — wire `RENDER_PASS_DIST` into `src/vis/rendererer.cpp`.